FFF Printing Fundamentals

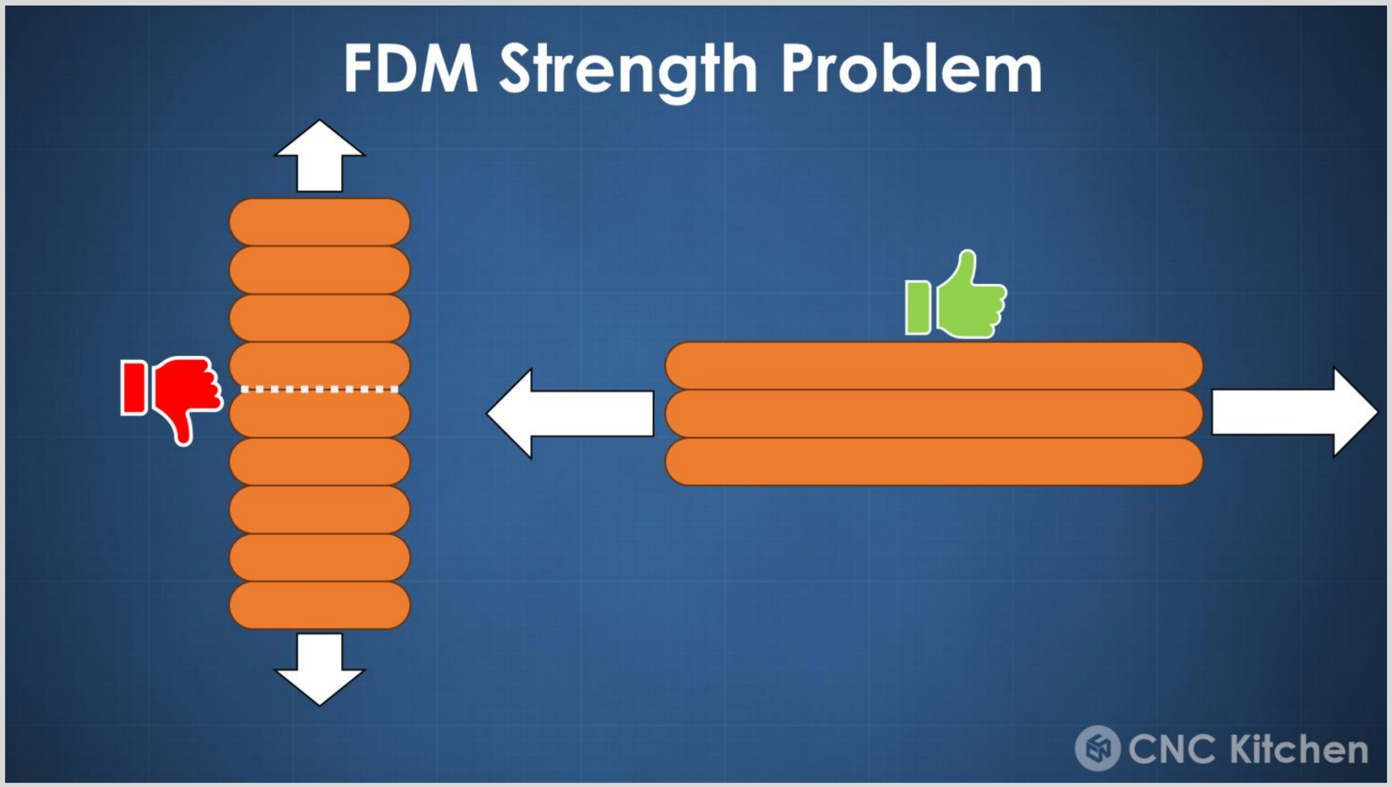
Part 2: Limitations

Thermal Cycle

- Some plastics shrink as they cool.
  - Up to 1 or 2%.
  - Seems small, but large enough to distort dimensions.
  - Leads to cracking and delamination in extreme cases.
- Advice?
  - Tightly control ambient temperature and cooling rate.
  - Some filaments (ABS, Nylon) like it hot → Enclosed print chamber, extra heating.
  - Some filaments (PLA) like it cold → Open-air printing, extra cooling.
  - PETG is somewhere in the middle.

Anisotropic Strength

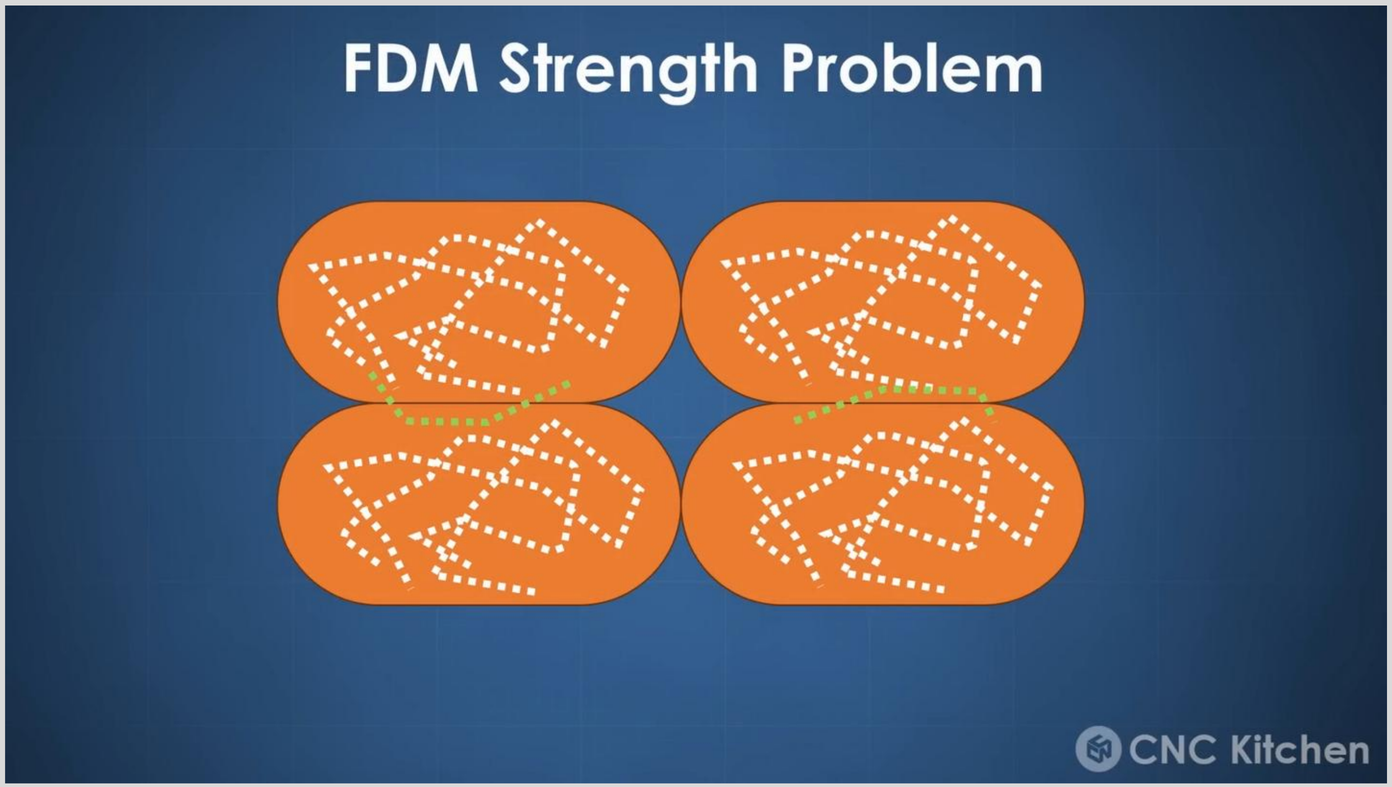
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- And those weak points align.



(YouTube video by CNC Kitchen)

Anisotropic Strength

- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament partially bonds to a cold layer.
- The “wrong” orientation can cost around 50% strength.



(YouTube video by CNC Kitchen)

Anisotropic Strength

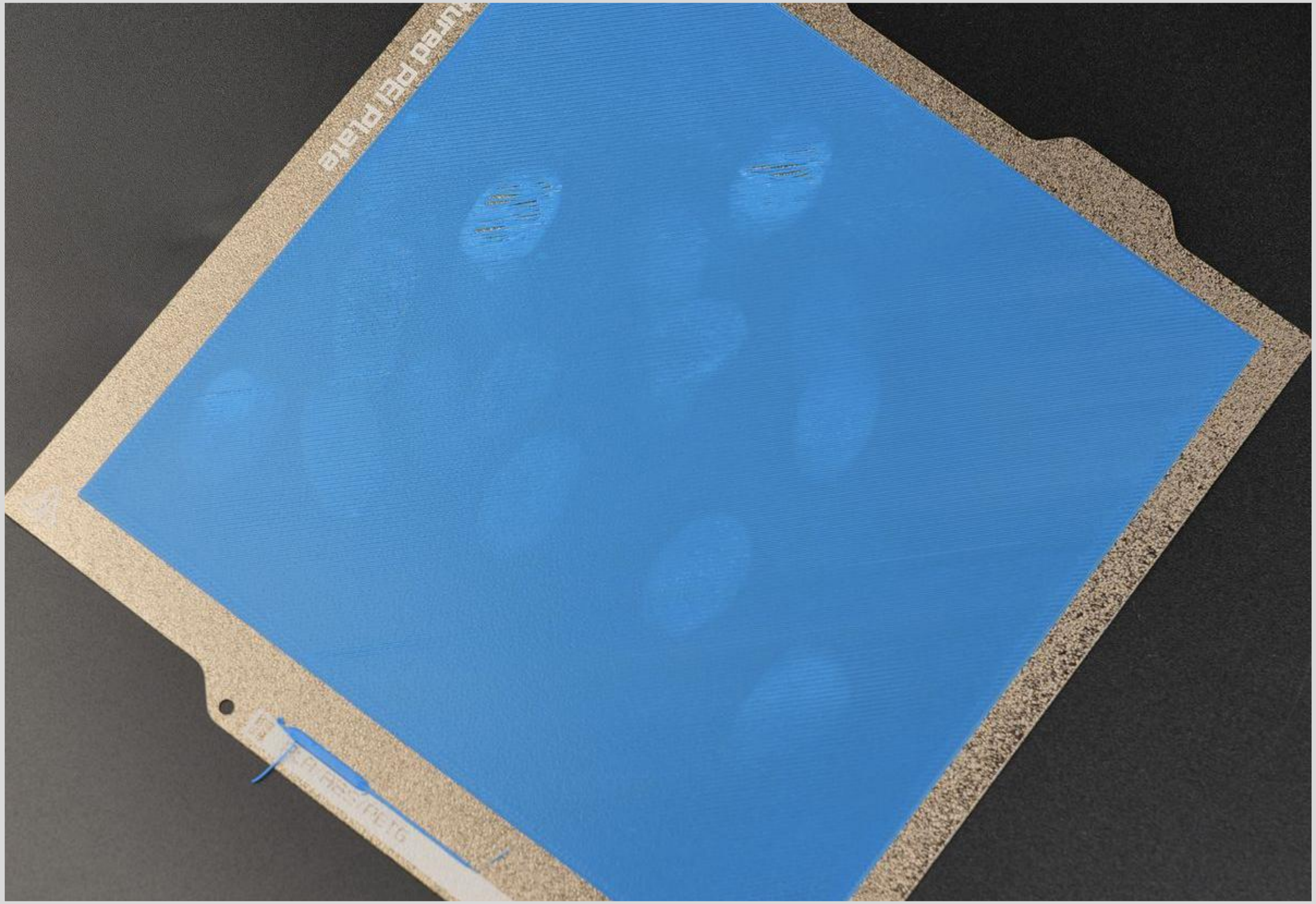
- Advice?
- Rotate prints to align layer lines and forces.
- Avoid tall and skinny shapes.
- Improve layer bonding.
  - Increase printing temps.
  - Decrease printing speed.

First Layer

- What is the most important layer in a 3D print?
  - The first layer!
  - First layers keep everything locked in place.
- Advice?
  - Always watch the first layer.
  - Tune first layer squish (extremely sensitive).
  - Flat surface area on the bottom of the 3D object.
  - Heated print bed.
  - Perfectly clean print bed.
  - Apply adhesion promoters (glue).

First Layer Mistakes

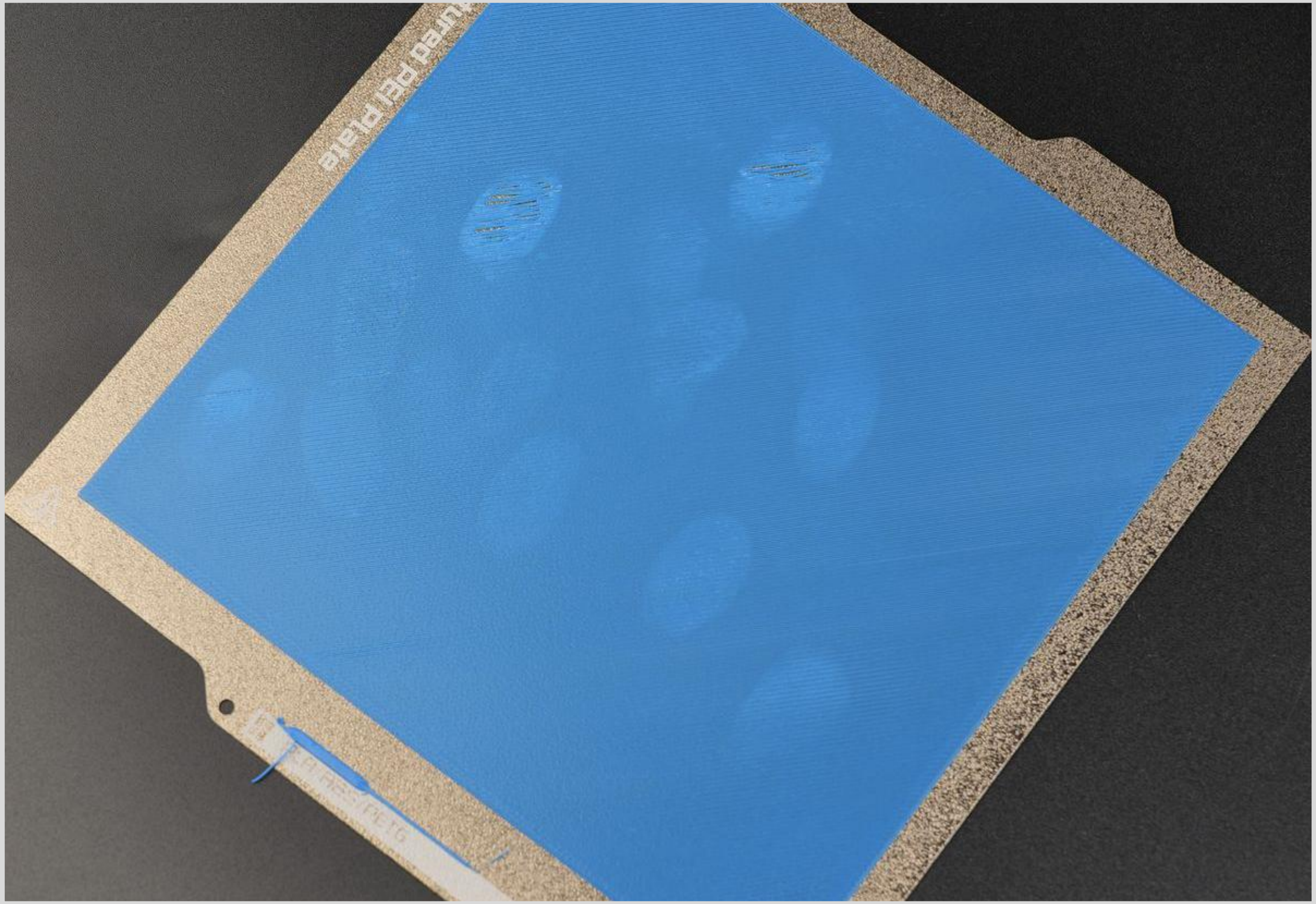
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from Bambu Labs Wiki)

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water at the sink.
- Everyday use: window cleaner with ammonia.



(Picture from Bambu Labs Wiki)

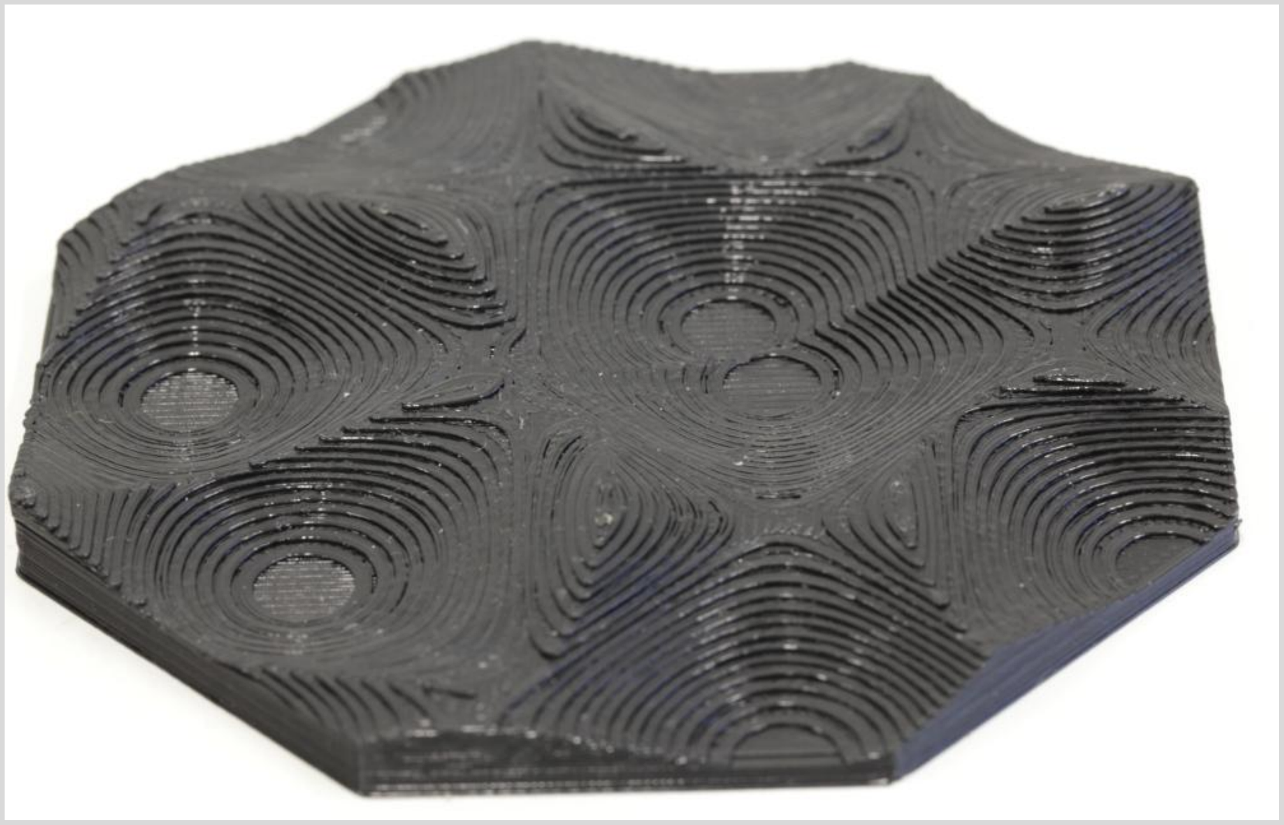
First Layer Mistakes

- What happens when a first layer does not stick?
- Print breaks free and makes "spaghetti".
- When did things go wrong?
- Look at the partially completed, rectangular print.

(Screenshot from Everything STEM)

Stair-Stepping

- Printing with layers →
  Vertical slopes turn into stair steps.
- Mostly cosmetic effect.
- More pronounced on shallow angles.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height at the cost of increased print time.



(Picture from Hamburg University Research Project)

First Layer Warping

- Why do corners and edges peel up?
- Regions shrink at different rates →
  Internal stresses →
  Stresses concentrate in corners.
- Shrinking is a percent →
  Large parts have the most problems.

(Picture from Creality Cloud Blog)

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.

(Picture from Creality Cloud Blog)

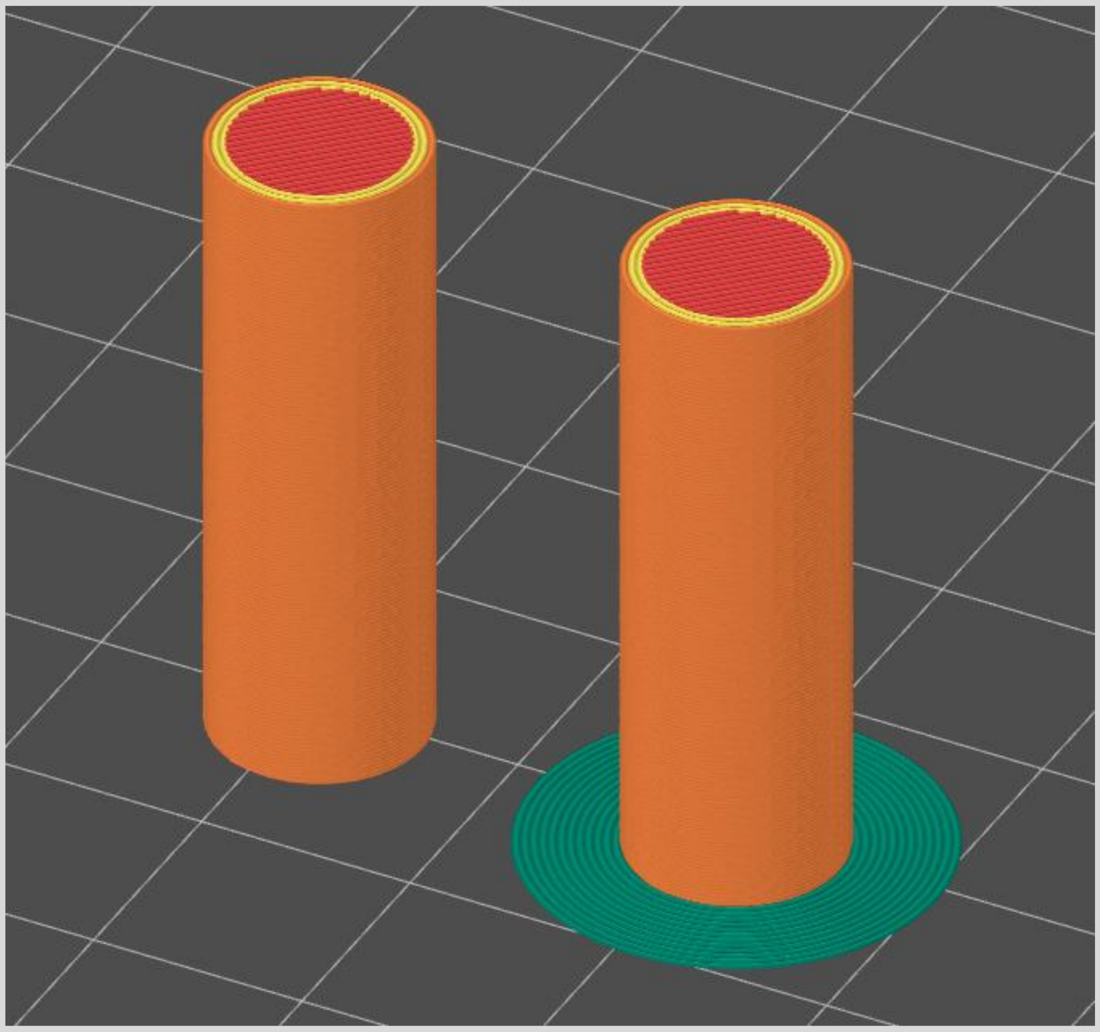
First Layer Warping

- Advice?
- Better control over temperatures.
  - Filament temperature.
  - Print chamber temperature.
  - Cooling fans.
- Evenly distributed cooling.
  - Avoid fans or drafts that cool one side.
- Round bottom corners to reduce stress concentration.

(Picture from Creality Cloud Blog)

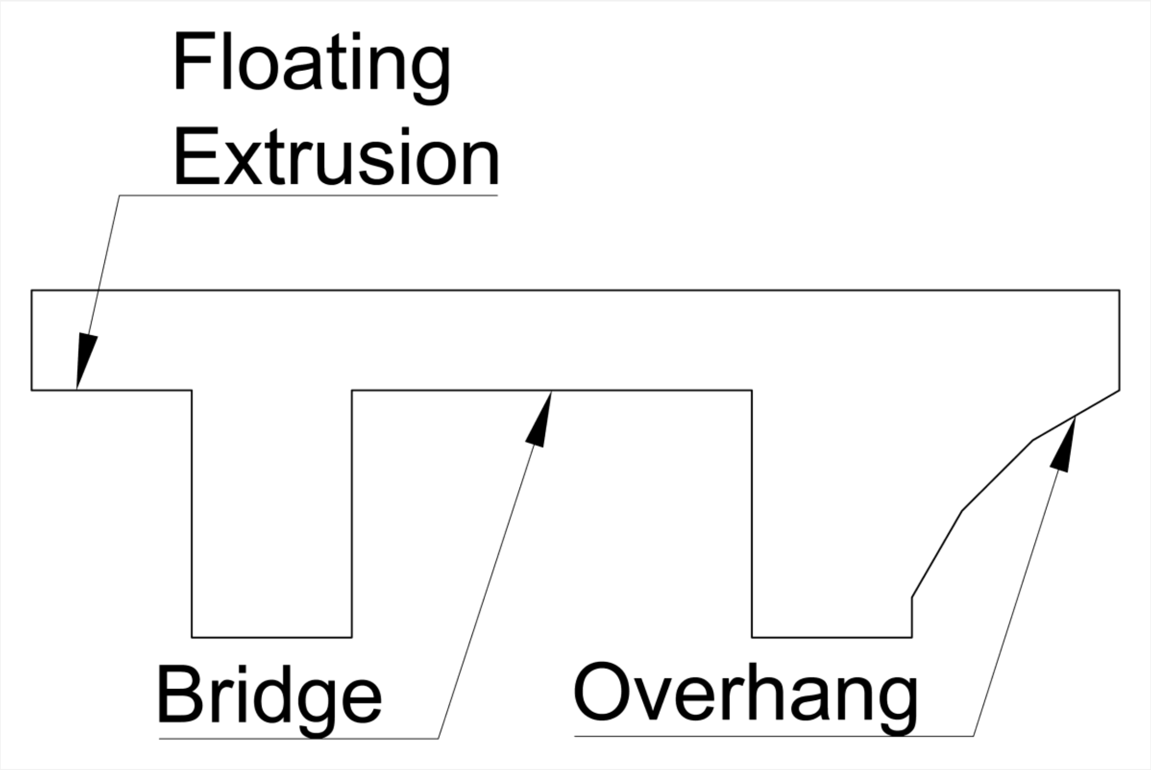
Brim

- What if you could add extra surface area to fight warping?
- Brim, like the brim on a hat.
- Concentric extrusions joined to bottom perimeter. (Green lines.)
- Added with a slicer.
- One layer tall.
- Easily peeled away after printing.



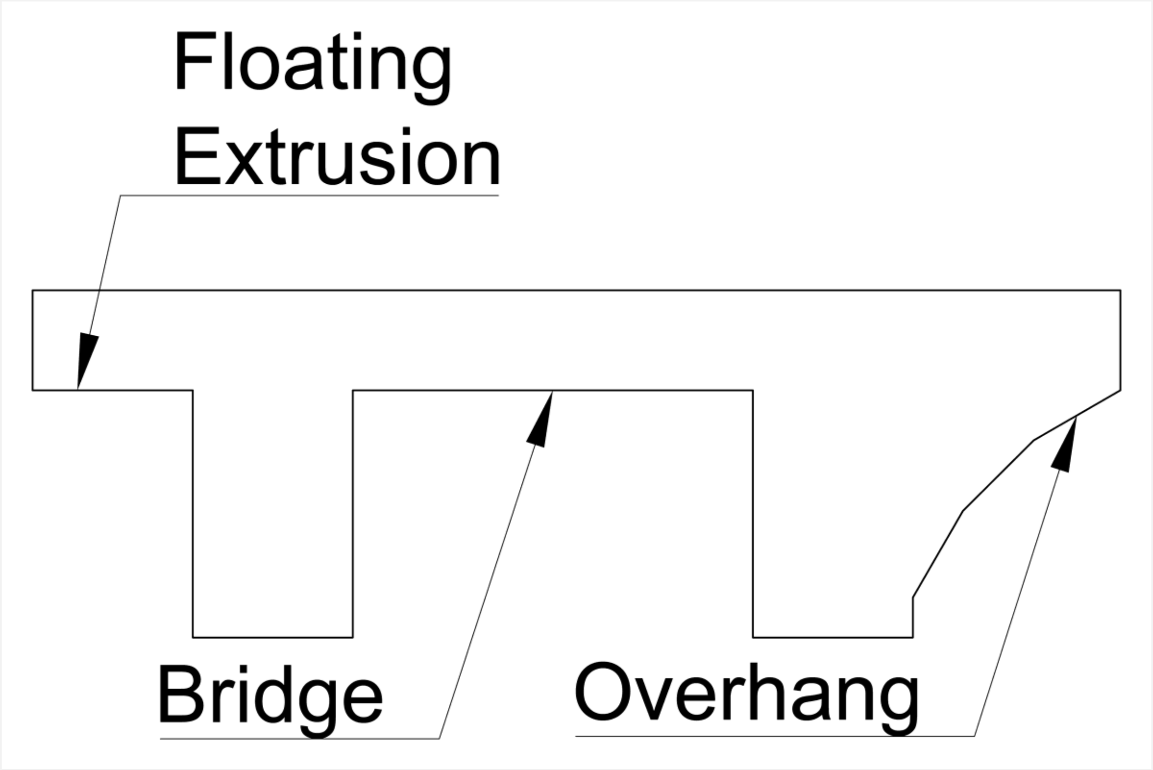
Levitating Extrusions

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



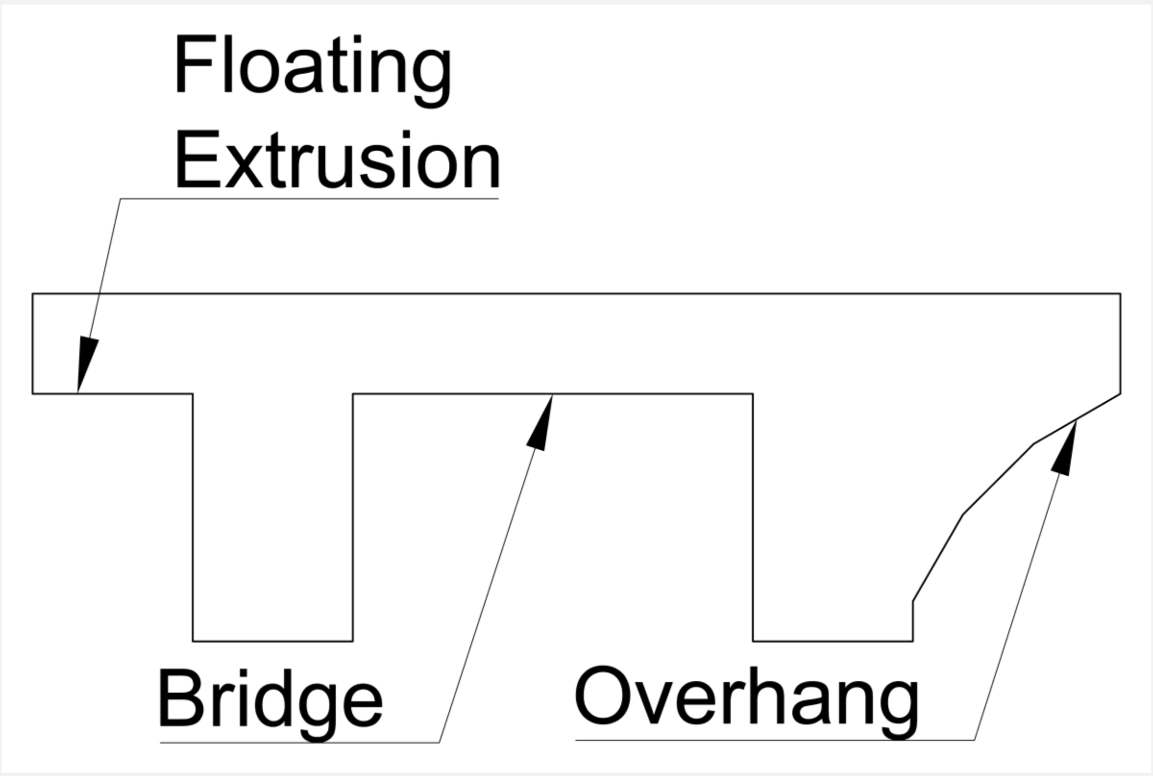
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



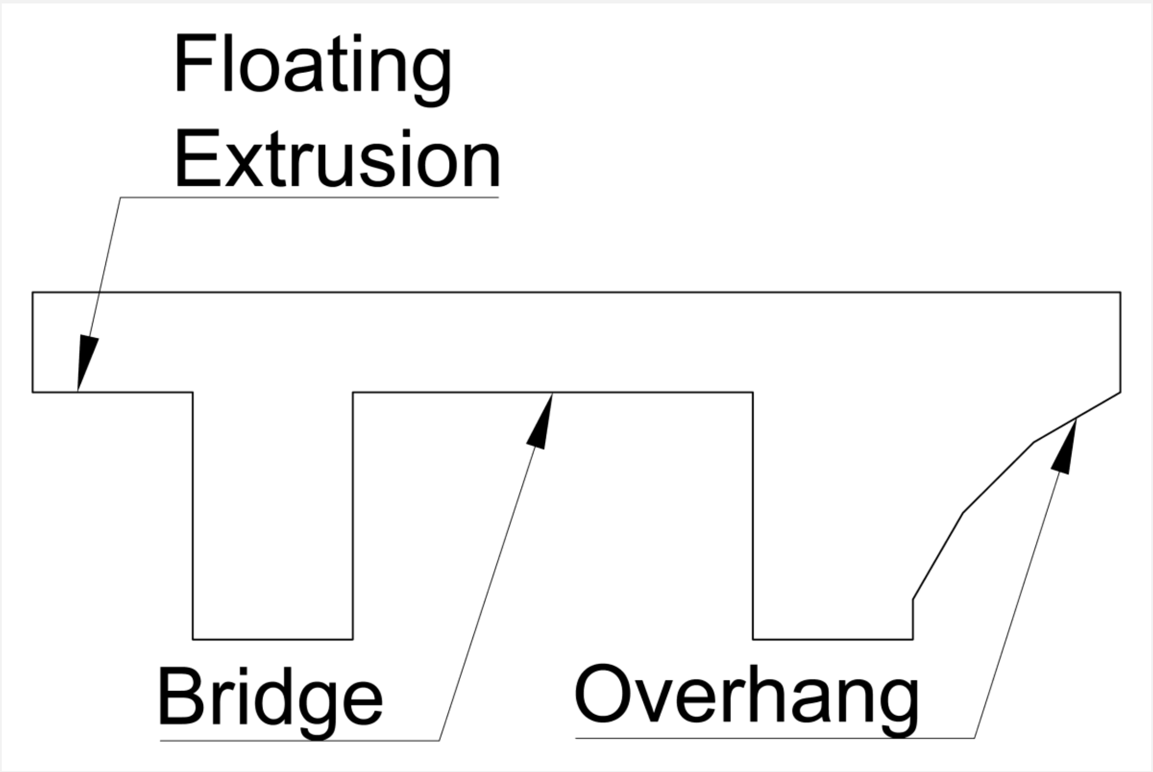
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- First few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



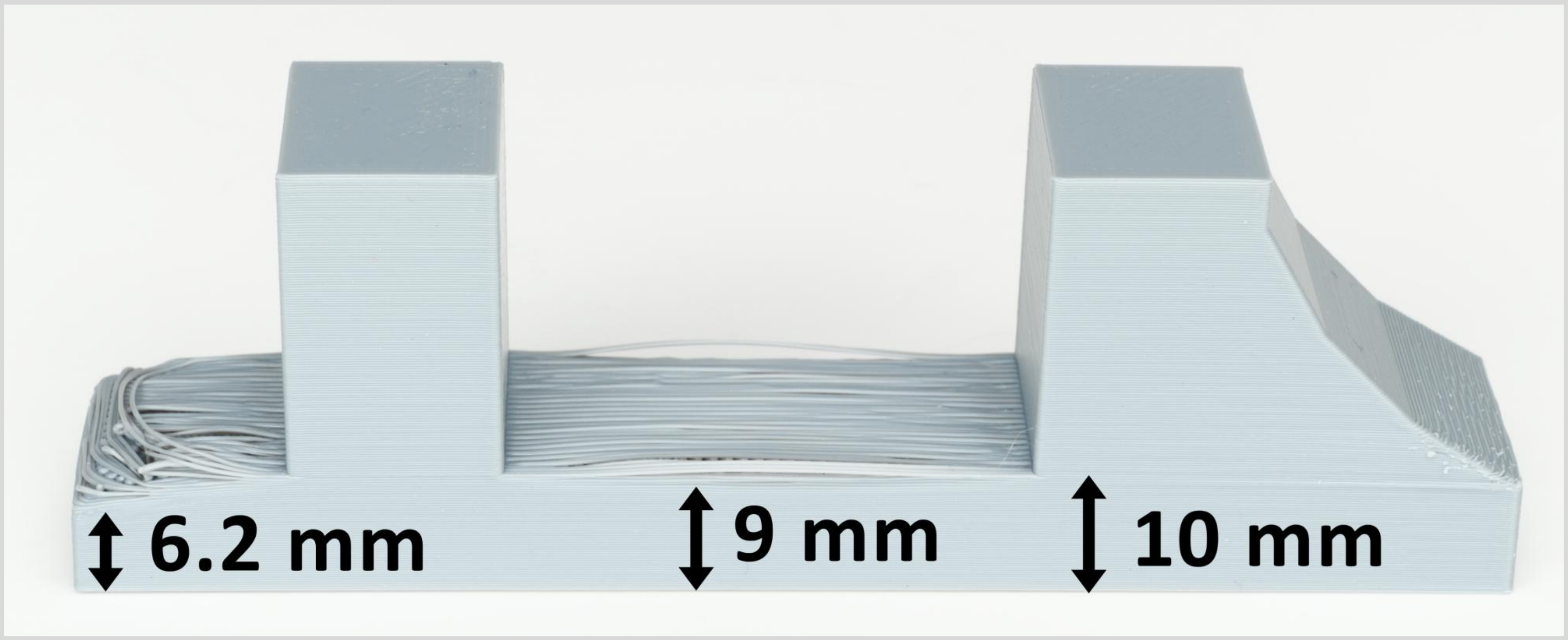
Floating Extrusion

- Extrusion that hangs in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



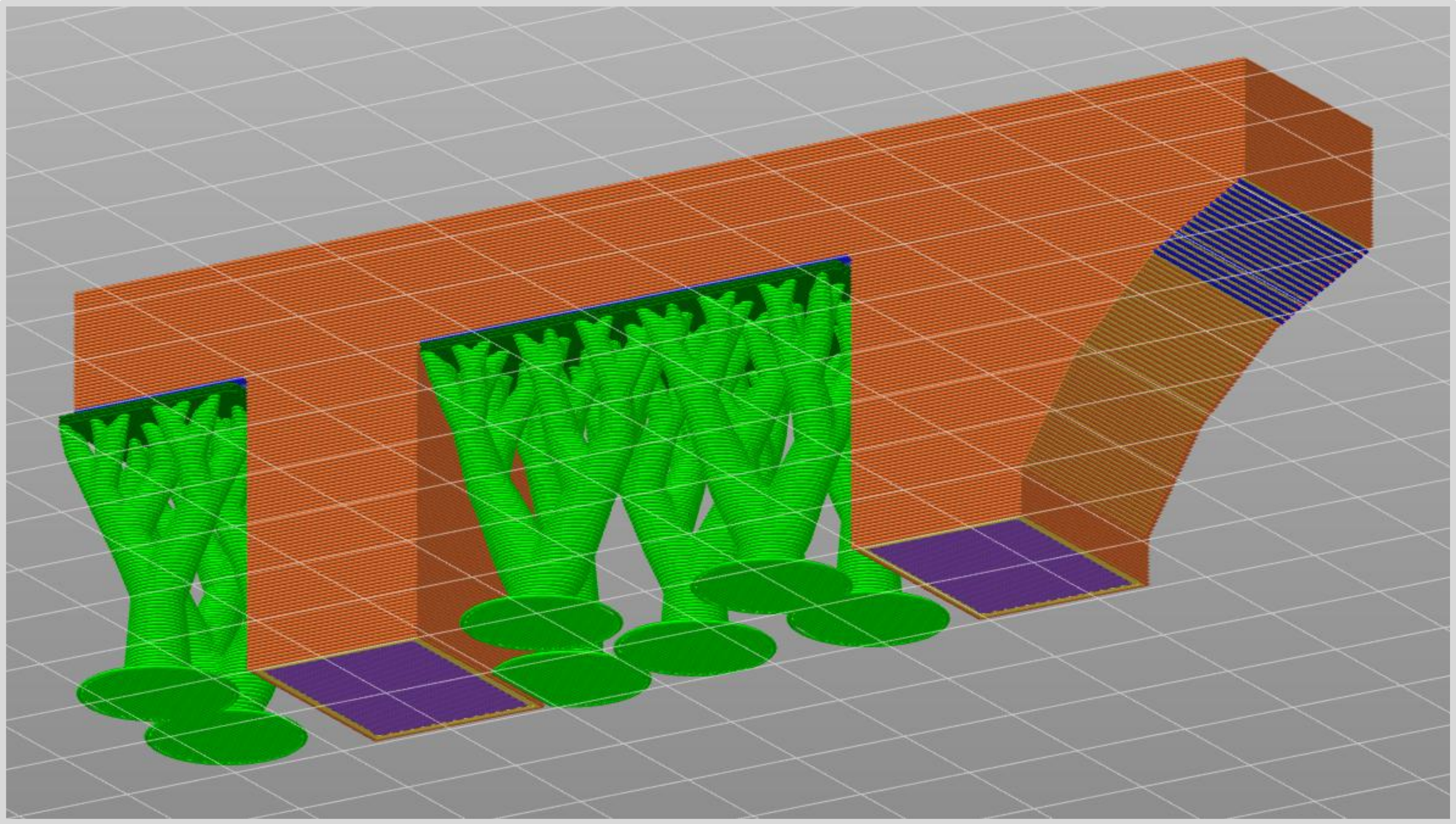
Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Temporary scaffold that fixes:
    - Floating extrusions.
    - Long bridges.
    - Steep overhangs.
- Added by the slicer. (Green trees-like branches.)
- Extra complexity, plastic, and print time.

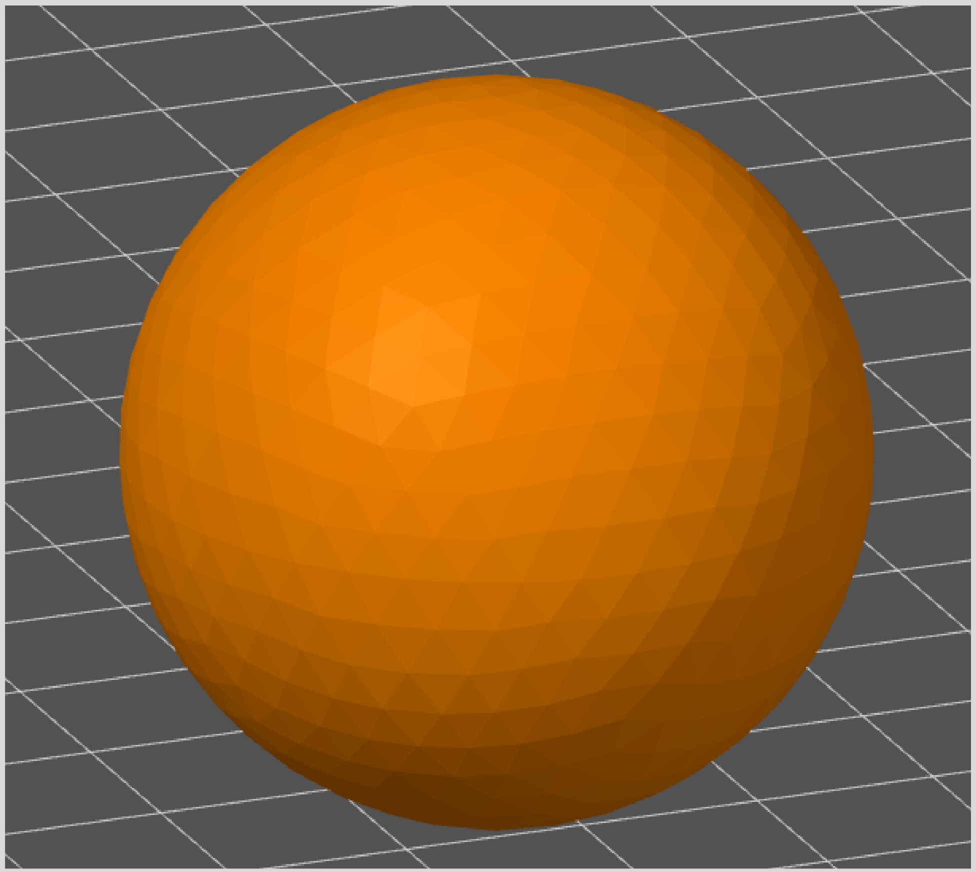


Supports

- In special situations, you can include supports as part of a 3D model.
  - *Example:* 3D Printing Trick by Clough42, and a follow-up video.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
  - If they do not bond well enough, they hold nothing in place.
  - If they bond too well, they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

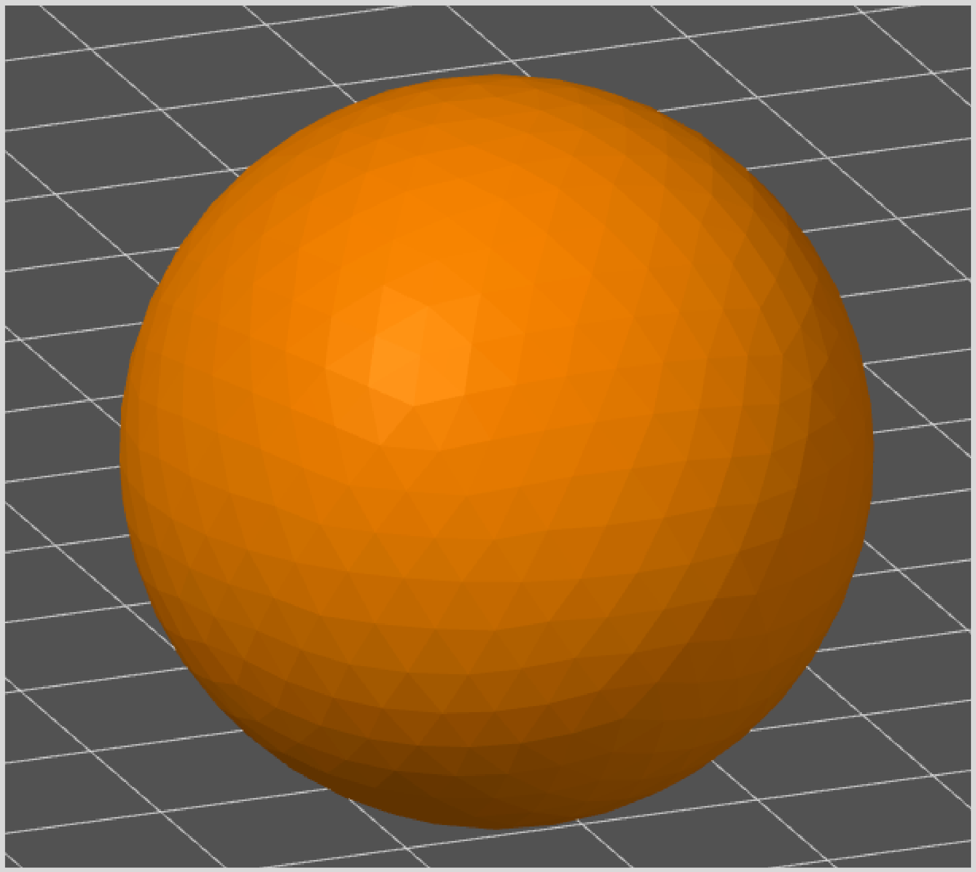
Limitations Review

- **Printability** summarizes limitations for a specific object.
- Without extra considerations, a simple sphere has poor printability.
- What limitations do you see?



Limitations Review

- Very bad first layer contact. Significant spaghetti risk.
- Steep overhangs on the bottom.
- Second layer is steep enough to create floating extrusions.
- Stair-stepping creates uneven surface finish.
- How can we fix this print?



Limitations Review

- Brim is not helpful.
  - Object starts with too little surface area.
- Add supports to the bottom.
- Cannot fix stair-stepping by changing print orientation.
  - Use small layer heights to reduce visual artifacts.
  - Use adaptive layer heights to make the surface look more uniform.
- Print two halves, then put them together.
  - Add tabs to help you align them.